Topic for Master 1/Master 2 internship

Academic year 2026-2027

From d = 4 to d = ∞ and back: taming strongly coupled quantum field theory with a 1/d expansion

Scientific description:

Context

Relativistic quantum field theory has been with us for about a century, yet we still cannot compute how particles interact, that is the S-matrix, beyond weak coupling. The prototypical example is QCD at low energies: quarks are confined into hadrons, which we can scatter in experiments, but whose collisions we cannot derive from first principles. Beyond weak coupling, we have very few tools.

The new idea of this internship is to look for a small parameter elsewhere: the number of spacetime dimensions d. The 1/d expansion works remarkably well in other areas of physics. In statistical mechanics, mean-field theory becomes exact at infinite d and 1/d gives systematic corrections; in general relativity, the large-d limit has given new analytic control over black holes. For scattering in quantum field theory, no such expansion exists yet, and the small parameter 1/d could unlock new universal features of strongly coupled interactions.

This project builds on our recent study of scattering in higher dimensions by the supervisor [1]. Using the S-matrix bootstrap, which bounds the space of possible interactions using only the consistency of relativity and quantum mechanics, we mapped using established optimization techniques which interactions are allowed in every dimension from 3 to 12, and observed surprising new phases appearing at special dimensions (5 and 7).

These results touch a deep question. Beyond six dimensions, no fully consistent interacting quantum field theory without gravity is known, and it has been conjectured that string theory might then be the only consistent option, an idea known as *string universality*. Yet our bounds do not seem to rule out high-dimensional interactions without gravitons. Is there really a theory without gravity behind them, or is a graviton hiding somewhere? This, and in general whether interacting theories without gravity exist in high dimensions, remains a completely open question.

Subject

The internship can go in two directions, analytic or numerical, or combine both, depending on the student's interests. On the analytic side, the student will study how the consistency conditions on the S-matrix simplify when d is large, and take first steps towards a 1/d expansion. On the numerical side, the student will use our code to push the computations to higher dimensions and test the large-d predictions.

The overall aim is to understand the phases of higher-dimensional quantum field theories and to look for traces of gravity hidden in them. A solid background in quantum field theory is expected; an interest in numerics, string theory or asymptotic methods is welcome.

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Team and environment

The student will join a small, friendly team of postdocs and PhD students working on these questions, sharing their office, with regular one-to-one supervision and weekly group meetings where questions of any level are welcome. The group runs a journal club and regular informal discussions, and attends the lab seminars. LAPTh is located in Annecy, between the lake and the mountains, close to CERN, and student housing is typically available for interns.

Opening towards a PhD thesis

The internship naturally opens to a PhD on quantum field theory and quantum gravity and string theory in higher dimensions. The thesis would develop a systematic 1/d expansion for particle collisions, turning the numerical phase diagram into an analytic understanding. It could then include gravitons explicitly, studying the S-matrix of quantum gravity in higher dimensions, as in [2], to test string universality directly and clarify where string theory sits in the space of consistent theories, as a unique option, or one amongst many others. The bootstrap philosophy is ideal for this: it is agnostic about string theory and relies only on first principles, so it can probe how quantum gravity works without assuming string theory from the start.

References

[1] M. Gumus, S. Metayer, P. Tourkine, *Tracking S-matrix bounds across dimensions* [arXiv:2512.24474]

[2] A. Guerrieri, J. Penedones, P. Vieira, *Where is String Theory?*, Phys. Rev. Lett. 127 (2021) 081601 [arXiv:2102.02847]

Possible extension as a PhD: YES

Keywords: S-matrix bootstrap, quantum field theory, higher dimensions, large-d expansion, string theory, quantum gravity, numerical optimisation

Supervisor(s):

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